Soft sensor: LPG bubble point / RVP-like proxy

Goal. Replace (or back up) a slow, maintenance-heavy analyzer with an online, model-based estimate of a hydrocarbon stream's volatility.

A common refinery/LPG-plant problem: an **RVP (Reid Vapor Pressure) analyzer** or gas chromatograph measures stream composition/volatility every few minutes at best, is expensive to maintain, and can fail or drift. If we already measure **temperature** continuously and can estimate or infer the stream **composition** (from a GC, a material balance, or an upstream soft sensor), we can compute the stream's **bubble-point pressure** at process/storage conditions using an equation of state (EOS). This tracks volatility in the same direction as RVP and can run every scan interval, with zero analyzer downtime.

This notebook builds that estimator using [Clapeyron.jl] with a cubic (Peng-Robinson) equation of state for a typical LPG mixture (propane / n-butane / isobutane / n-pentane).

> This is a teaching/demo report with representative, not plant-specific, data, evaluated at a fixed operating point (the interactive notebook version lets you adjust composition and temperature with sliders).

2. Operating point used in this printout

Component	Mole fraction
Propane	0.4
n-Butane	0.35
Isobutane	0.15
n-Pentane	0.1

Temperature: **40.0 °C**

```
components = ["propane", "butane", "isobutane", "pentane"]
model = PR(components)
```

3. Soft sensor output: inferred bubble-point pressure

Inferred bubble-point pressure (soft-sensor output): 735.7 kPa (106.7 psi)

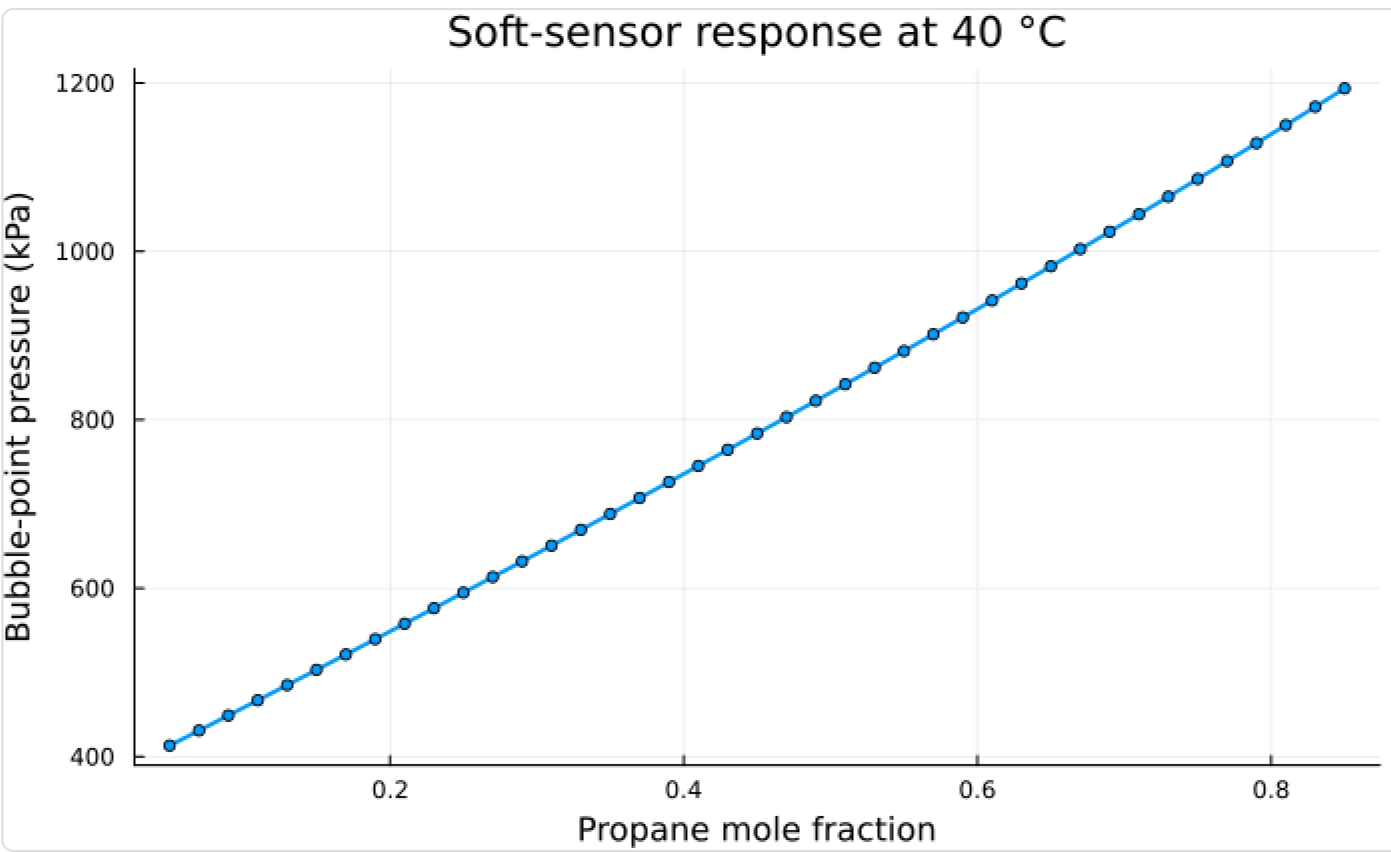
Higher bubble-point pressure at a given temperature ⇒ more volatile stream (more propane/butane) ⇒ higher RVP. In a real deployment this value would be trended against the periodic lab RVP result and, if needed, corrected with a simple bias/trim term.

```
p_bubble, vl, vv, y = bubble_pressure(model, T, x)
p_bubble_kpa = p_bubble / 1000
```

Bubble-point pressure ≈ 735.7 kPa (106.7 psi) at 40.0 °C

4. Sensitivity: how the soft sensor responds to composition

Sweep the propane fraction (renormalizing the rest proportionally) at a fixed temperature to see how the inferred bubble-point pressure — our RVP proxy — responds. A soft sensor should move monotonically and smoothly with the property it's inferring; this kind of check is a basic sanity test before trusting it against plant data.



5. Next steps toward a deployable soft sensor

- Replace the interactive sliders with a live/historized composition feed (GC, material balance, or an upstream inferential estimate) and temperature tag from the plant historian.
- Validate the EOS-based estimate against periodic lab RVP samples over a representative operating range; fit a simple bias/trim correction if a consistent offset appears.
- Consider switching the EOS (e.g. SRK, or an activity-coefficient model via Clapeyron) if validation shows the cubic EOS underperforms for this specific stream.
- Package the estimator as a function in `src/` so it can be called from a scheduled job or OPC/historian integration, independent of the notebook.

Generated from notebooks/01_lpg_rvp_soft_sensor.jl (Clapeyron.jl PR EOS) — caxelrud/Soft-sensor